

BZAN 535: Team Project 1

Private Label at Kroger

Presentations: Thursday, October 1, 2026

The Situation

Private label, the store's own brand, is one of the most valuable levers a grocer has. Private-label products typically earn the retailer a higher *percentage* margin than the national brands sitting beside them on the shelf, and a shopper who trusts the store brand is a shopper who has a reason to come back. But private label also competes for finite shelf space, and pushing it in the wrong aisle, or at the wrong customer, costs a retailer both the sale and some goodwill.

Kroger's merchandising team has two years of loyalty-card transaction data and a partial demographic profile of its households. They have asked your team a deliberately open question:

Who buys our private-label products, where do they buy them, and what should we do with that information?

They have not told you which customers to look at, which categories matter, or what “buys private label” should mean. Those are your decisions, and defending them is a substantial part of this assignment.

Your Charge

Deliver a short, evidence-based recommendation to the merchandising team. A good project takes a position — *this* segment, *this* part of the store, *this* action — and supports it with analysis the team could actually act on. A weak project reports every number it computed and leaves the audience to work out what matters.

You are not expected to answer the whole question. You are expected to answer *a* well-chosen piece of it thoroughly, and to be honest about what your evidence does and does not support.

The Data

Three files, all on Canvas:

- `transaction_data.csv` — roughly 2.6 million rows. One row is one product in one basket. Includes `household_key`, `BASKET_ID`, `PRODUCT_ID`, `SALES_VALUE`, `QUANTITY`, `STORE_ID`, `WEEK_NO`, `DAY`, `TRANS_TIME`, and `discount` fields.

- `product.csv` — one row per product. Includes `BRAND` (National or Private), `DEPARTMENT`, `COMMODITY_DESC`, and package size.
- `hh_demographic.csv` — demographic profile for a *subset* of households: income, age, home-owner status, marital status, household size, composition, and children.

The files join on `PRODUCT_ID` and `household_key`.

What Your Analysis Must Demonstrate

These are requirements about *rigor*, not about which question you ask.

1. **A defined and defended unit of analysis.** State plainly what one observation is in your analysis — a single product line, a basket, a household, a store — and why that is the right choice for the question you asked. This single decision changes your answer, and we will ask about it.
2. **Description first.** Show us the distribution of whatever you are going to make claims about. At least one figure or table should earn its place by changing what the audience understands.
3. **Methods that fit the question.** Draw on what we have covered — descriptive summaries and figures, confidence intervals, comparisons of two or of several groups, tests of association — and choose among them on the basis of what your question actually needs. A well-built table can settle a question that needs no test at all. Equally, a difference you want to call real needs more than a gap between two averages.
4. **Uncertainty and magnitude.** Wherever you claim a difference, tell us how large it is and how sure you are of it, in business language rather than statistical boilerplate. If you ran several comparisons, say how many and account for it.
5. **An interrogation of your own finding.** Check the assumptions your method depends on. Consider at least one alternative explanation for the pattern you found, and tell us whether you can rule it out.
6. **A clear statement of limits.** What would you need — different data, a different design — to make a stronger claim than the one you are making?

Choosing Your Angle

Teams will be graded partly on the quality of their framing, so pick something you can defend. “Where” is deliberately open — it can mean which aisle or which location. Three natural dimensions to vary:

- **Which customers.** The demographic file offers several ways to slice households. They are not equally informative, and finding out which ones carry signal is part of the work.
- **Which part of the store.** Private label does not behave the same way in every department. A finding that holds overall may not hold in the aisle you care about — and that discrepancy is often the interesting result.

- **Which store.** The data cover hundreds of locations, and they do not all sell the same mix. Characterising how they differ, and what those differences reflect, is part of the work.

You are also free to define “buys private label” however you can defend: share of dollars, share of items, share of trips containing one, or something else. These are not interchangeable, and they may not all give you the same answer. Choose the one that matches the question you are asking, and tell us why you chose it.

Deliverable

An **8-minute presentation** on Thursday, October 1, followed by about 3 minutes of questions. Every team member presents some portion.

Submit your slides and your `.Rmd` (or script) to Canvas before class. Code will not be graded line by line, but it must reproduce the numbers on your slides.

Roughly six to eight slides is the right scale. Do not build an appendix of everything you tried; build the argument you want to make.

Grading

Worth **10% of the course grade** — 5% content, 5% presentation.

Content	50
Framing — a specific, well-motivated question with a clear business rationale	5
Unit of analysis — correctly chosen, explicitly justified, consistently applied; sensible data handling decisions	10
Descriptive work — distributions shown, figures that inform rather than decorate	10
Evidence — methods suited to the question, correctly executed; uncertainty and magnitude reported; multiplicity handled where relevant	15
Limitations — assumptions checked, alternative explanations considered, claims matched to evidence	10
Presentation	50
Narrative — the argument builds; the recommendation is clear and lands	15
Slides — readable, uncluttered, labeled axes, no raw console output	10
Delivery — within time, well paced, not read from the screen	10
Questions — answers directly, concedes uncertainty honestly	10
Participation — the work is visibly shared across the team	5
Total	100

Individual Contribution

The project grade is a team grade, but it is not automatically the same for everyone. Within 24 hours of the presentations, each of you will complete a short confidential peer evaluation — a Google Form linked from Canvas — rating your teammates on their share of the analysis, their share of the write-up, their reliability, and their engagement with the team's thinking.

Responses go only to the instructor.

Most of you will likely receive no adjustment. The evaluation is scored so that ordinary differences of opinion leave the team grade untouched; it moves a grade only when teammates agree, clearly, that someone contributed substantially more or less than a fair share. A grade can be adjusted up or down.

Completing the form is required.